YUDONG LUO

(412) 616-6226 | yudongluo.cmu@gmail.com | www.linkedin.com/in/yudong-luo

EDUCATION

Carnegie Mellon University Dec 2026
Master of Science in Artificial Intelligence Engineering - Mechanical Engineering
Concentration: **Robotic and Control Systems**
Selected Coursework: Robot Learning; Computer Vision; Multimodal Machine Learning; Trustworthy AI; Deep Learning

University of Electronic Science and Technology of China (UESTC) Jun 2025
Bachelor of Engineering in Communications Engineering

SKILLS

Languages: Python, C/C++, MATLAB
Robotics & Simulation: ROS, Isaac Sim/Lab, MuJoCo, PCL, Open3D, OpenCV
ML & Tools: PyTorch, CUDA, Docker, Linux, Git

PROJECTS & RESEARCH EXPERIENCE

EgoX: Egocentric Cross-Embodiment Learning for Versatile Manipulation | *CMU × Bosch Research* Apr 2026 – Present
Safe AI Lab | IROS 2026 Workshop (outstanding paper award)

- Implemented the modality tokenizers and shared encoder-decoder trunk: a shared visual backbone for image streams and lightweight MLPs for state modalities, both compressed by cross-attention with learnable queries into a fixed token budget, exposing one interface across embodiments.
- Developed the R1 Lite integration, controller, and data-collection pipeline, and built rollout and evaluation workflows; on xArm, cross-embodiment pretraining raised Scoop success from 61.5% to 84.6% and OOD Sort from 1/7 to 5/7.

Skill-Specialized Diffusion Policy with Task-Space Residual RL | *CMU* May 2026 – Present
IROS 2026 Robotic Assembly Competition, RoCo Task Board | Team of 3

- Designed a skill-specialized diffusion policy for 9-part sequential assembly: a shared visual-state encoder feeds three DDPM action heads grouped by end-phase motion, FiLM-conditioned on part type, pick-place coordinates, and release mode.
- Formulated a task-space residual RL layer applied in two stages, at grasp and at placement, for mm-tolerance parts: a lightweight policy outputs a clipped 6-DoF Cartesian correction mapped to joint space via a damped Jacobian pseudo-inverse, pretrained on demonstration residuals and finetuned with TD3 in Isaac Sim.
- Benchmarked task-grouped π0.5 against a task-grouped diffusion policy as backbones; the residual layer transferred better onto the diffusion policy, lifting the best-performing part from 61% to 91% grasp success and triggering snap insertion at 3.79 mm / 2.95°.

Semantic-Aware Robotic Grasping from Human Egocentric Videos | *CMU* Jan 2026 – Jun 2026

- Built an egocentric hand-object interaction pipeline: 3D reconstruction with MoGe2, hand pose in MANO parameter space with HoWoR, plus contact-point and object-pose estimation; auto-labeled segmentation and grasps with Qwen3 VLM+SAM3.
- Designed a decoupled human-to-robot mapping: fitted a kernel density estimate over contact points and approach poses extracted from human demonstrations to form a per-object grasp prior, then used it to re-rank Contact-GraspNet candidates in gripper configuration space, improving grasp success by up to 20% on the best-performing object.
- Built a three-layer pipeline — Perception (RealSense D405 → Moondream2 VLM → EdgeTAM) → Planning (Contact-GraspNet) → Execution (IK → trajectory) — fronted by a conversational agent that turns free-form user requests into grounded targets, reaching 83% success across 5 object types.

Generative Robot Policy Optimization with Diffusion & Flow Matching | CMU Sep 2025 – Dec 2025

- Benchmarked Flow Matching against Diffusion Policy on Push-T with ODE-based sampling; Beta-distribution timestep sampling raised single-step-inference success from 0.13 to 0.82, cutting latency for real-time control.
- Investigated online policy adaptation with Diffusion Steering RL (DSRL), running ablations on sparse-reward dynamics and tuning hyperparameters to improve training stability by ~25%.

Learning-based Multi-Camera Visual Odometry | *CMU* Sep 2025 – Mar 2026
Research Assistant, AirLab × Denso

- Generalized MAC-VO to arbitrary N-camera rigs via configurable camera-pair matching and cross-view triangulation, unifying pretrained optical flow networks (VGGT, UFM, RoMaV2) into one uncertainty-aware frontend.
- Implemented segmentation-based motion masking to filter dynamic objects and used the masks to train a keypoint selector, improving VO robustness in dynamic scenes.

PROFESSIONAL EXPERIENCE

HiGaAs Microwave Technology Co., Ltd. Jul 2024 – Aug 2024
Testing and Automation Intern, System-level Packaging Product Department *Chengdu, China*

- Built an automated instrument-control and data-acquisition system for chip reliability testing, driving benchtop equipment over SCPI/Socket and automating temperature cycling, logging, and analysis.

PUBLICATION & PATENT

Egocentric Cross-Embodiment Manipulation with Embodiment Dreaming. IROS 2026 Workshop (outstanding paper award). Full paper under review, ICLR 2027.

Qin, L.; Luo, Y. *Nighttime Traffic Scene Salient Object Detection with Improved YOLOv9.* CN App. 2024119132237, 2024. Patent Pending.